

The Art of Two-Round Voting

Qishen Han

Rutgers University
New Brunswick, NJ, United States
hnickc2017@gmail.com

Biaoshuai Tao

Shanghai Jiao Tong University
Shanghai, China
bstao@sjtu.edu.cn

Grant Schoenebeck

University of Michigan
Ann Arbor, MI, United States
schoeneb@umich.edu

Lirong Xia

Rutgers University
New Brunswick, NJ, United States
lirong.xia@rutgers.edu

Abstract

We study the voting problem with two alternatives where voters' preferences depend on a not-directly-observable state variable. While equilibria in the one-round voting mechanisms lead to a good decision, they are usually hard to compute and follow. We consider the two-round voting mechanism where the first round serves as a polling stage and the winning alternative only depends on the outcome of the second round. We show that the two-round voting mechanism is a powerful tool for making collective decisions. Firstly, every (approximated) equilibrium in the two-round voting mechanisms (asymptotically) leads to the decision preferred by the majority as if the state of the world were revealed to the voters. Moreover, there exist natural equilibria in the two-round game following intuitive behaviors such as informative voting, sincere voting [2], and surprisingly popular strategies [27]. This sharply contrasts with the one-round voting mechanisms in the previous literature, where no simple equilibrium is known. Finally, we show that every equilibrium in the standard one-round majority vote mechanism gives an equilibrium in the two-round mechanisms that is not more complicated. Therefore, the two-round voting mechanism provides a natural equilibrium in every instance, including those where one-round voting fails, and it can reach an informed majority decision whenever one-round voting can. Our experiments on LLM voters also imply that two-round voting leads to the correct outcome more often than one-round voting under some circumstances.

CCS Concepts

• Theory of computation → Algorithmic game theory.

Keywords

Computational Social Choice, Algorithmic Game Theory

ACM Reference Format:

Qishen Han, Grant Schoenebeck, Biaoshuai Tao, and Lirong Xia. 2026. The Art of Two-Round Voting. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3774904.3792275>



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792275>

1 Introduction

How can people make wise informed collective decisions? In many scenarios, people's preferences depend on the information they receive, and a collective decision needs to be made fairly. For example, suppose that a community wants to decide whether or not to adopt a restrictive COVID-19 policy. The preferences of the community members depend on their perceptions of the virality of the pandemic. While the virality is not directly observable, community members receive (noisy) signals about it to form their preferences, and their preferences can still be different even upon receiving the same signal. For example, immunocompromised people may still prefer to adopt the restrictive policy when the signal is "moderately dangerous", while healthy people may prefer not to adopt the policy after receiving the same signal. Other examples include political elections, recruiting, group activity selection, etc.

A natural idea is to use voting to make collective decisions. When there are two alternatives, majority voting appears to be a natural choice. The Condorcet Jury Theorem [10] establishes that when individuals receive independent signals about an unknown state of the world (such as the virality of COVID-19 in the above example) and vote honestly as their signals reflect, majority voting reveals the true state with almost surely as the population grows large. Moreover, when voters are strategic, majority voting can still aggregate voters' fully informed preferences via strategic behaviors [15, 21]. In other words, it chooses the majority winner *as if the state of the world were revealed to the voters*. In the literature, such a majority decision has been referred to as *full information equivalence* [15], *majority wish* [28], and *informed majority decision* [21]. In this paper, we will use the term "informed majority decision".

However, all equilibria of such one-round majority games in previous work are quite complicated and hard to follow. For example, the seminal paper by Austen-Smith and Banks [2] shows that, under majority voting, the natural and intuitive behavior of *informative voting*, in which every voter votes for his/her signal, and *sincere voting*, in which every voter votes as if he/she is making an individual decision and vote for his/her preferred alternative in expectation, are not Nash equilibria in most situations. The equilibrium constructed by Feddersen and Pesendorfer [15] requires the

The full version of this paper, including the Appendix and all the proofs, is at <https://arxiv.org/abs/2505.10377>.

Grant Schoenebeck gratefully acknowledges the support of NSF awards #2313137 and #2208662. The research of Biaoshuai Tao was supported by the National Natural Science Foundation of China (No. 62472271). Lirong Xia acknowledges NSF #2450124, #2517733, and #2518373 for support.

computation of a complicated Riemann integral. The equilibrium by Han et al. [21] involves carefully calculated mixed strategies.

In analyzing strategic behavior, we emphasize the importance of natural and intuitive equilibria. It is quite unlikely that real-world agents can calculate out and play counterintuitive mixed or asymmetric strategies, let alone coordinate to the same equilibrium. Therefore, these counterintuitive strategy profiles fail to predict the outcome of strategy agents as they are intended to. Moreover, the simplicity of the mechanism is also a desirable property and a proxy of usability, as the theoretical guarantees of complex mechanisms may not translate into practice.

Can we design simple mechanisms with natural equilibria to reach the informed majority decision?

The definition of a natural equilibrium depends on the scenarios and the purposes. In this paper, we identify the following criteria. Firstly, agents play pure/deterministic strategies. Mixed or randomized strategies are hard for agents to compute and implement and are rarely applied by real-world voters. Secondly, agents play symmetrically, i.e., agents with the same preferences should play the same strategy. Otherwise, the mapping from similar agents to diverse strategies is unclear and unincentivized, and the equilibrium is unlikely to predict the outcome. Moreover, agents can easily compute the strategies. Two examples of natural strategies are informative voting and sincere voting, where the first simply follows the signal, and the second follows the preferences in expectation.

Unfortunately, previous one-round voting mechanisms fail to have a natural equilibrium. Moreover, **one round voting game with the majority rule has NO pure symmetric equilibrium** under a large class of games (Proposition 1). The known equilibria are either randomized or asymmetric and are hard to compute.

EXAMPLE 1 (FAILURE OF INFORMATIVE VOTING BEING A NASH EQUILIBRIUM [2, 21]). Consider the aforementioned COVID-19 policy problem. Agents are deciding whether to accept or reject a restrictive policy. The world state, a.k.a. the virality of the pandemic, can be of High risk or Low risk with an equal prior probability. Agents receive High or Low signals related to the world state. The signal distribution under different risk levels and the utility of agents under different states and outcomes are specified in Table 1. The signal distribution is biased, as an agent is always more likely to receive a High signal, no matter what the risk level is.

Table 1: Informative voting fails to be a Nash equilibrium.

(a) Signal distributions.		
State of the World	High Signal	Low Signal
High Risk	0.9	0.1
Low Risk	0.6	0.4

(b) Agents' utilities.		
State of the World	Accept	Reject
High Risk	1	0
Low Risk	0	1

A strategic agent considering a unilateral deviation only cares about the case where exactly half of the other agents vote for Accept

and the other half vote for Reject because otherwise her vote will not affect the outcome. If all other agents vote informatively, this implies exactly half of the other agents receive High signals. However, agents receive High signals with a high probability of 0.9 when the risk level is High. Therefore, the world state is much more likely to be Low risk when half of the agents receive Low signals, and a strategic agent will deviate and vote for Reject even if her signal is High. This implies that informative voting fails to be an equilibrium.

Another line of work designs mechanisms to achieve the informed majority decision via simple equilibria, yet the mechanism structures are sophisticated. Schoenebeck and Tao [28] propose an information-aggregation mechanism that incentivizes truthful reporting and reaches the informed majority decision with a high probability. However, the mechanism leverages the “surprisingly popular” mechanism [27] and the median-voter theorem [8], resulting in a structure that is quite sophisticated and may not be trusted by the people. Moreover, the mechanism elicits a distribution from each agent, which requires infinite bits to be represented. This further increases the difficulty of implementing the mechanism.

1.1 Our Contribution

In this paper, we provide a solution to the research question of simple mechanism design—*two-round voting*, which has a polling stage (first round) and a voting stage (second round). Between the two rounds, agents can observe the number of votes cast for each alternative in the first round, and the majority vote in the second round completely determines the winner.

For a sharp contrast, we first show that the one-round voting game with the majority rule has NO pure and symmetric equilibrium under a large class of games (Proposition 1), which also rules out the natural informative voting and sincere voting. We also illustrate that the existing mixed or asymmetric equilibria bring difficulties for agents to calculate and coordinate.

On the other hand, the two-round voting mechanism is a simple and powerful mechanism with natural equilibria. Firstly, we prove that the two-round voting mechanism indeed achieves good decisions with strategic agents, as every ϵ -approximate strong Bayes Nash equilibrium, in which no group of agents has incentives to deviate, leads to the informed majority decision with high probability (Theorem 1). The probability converges to 1 as ϵ goes to 0. Secondly, the two-round voting mechanism has the merit of a simple mechanism in both structure-wise and equilibrium-wise. It has a simple structure that involves simply conducting a poll before the actual vote. Such a polling-voting structure has been widely applied in real-world elections.

Moreover, we show a natural ϵ -approximate strong Bayes-Nash equilibrium that leads to the informed majority decision (Theorem 2). As the number of voters increases, ϵ converges to 0. The equilibrium combines the most natural and classical informative voting and sincere voting: agents whose preferred alternative changes with the world state vote informatively (vote for their signal) in the first round, and vote sincerely (vote for the alternative they preferred in expectation conditioned on the first-round outcome) in the second round. The “informative + sincere” equilibrium follows the intuition of sharing information and choosing the (expected) preferred

alternative. We further discover that the “informative+sincere” equilibrium belongs to a class of “informative + threshold” equilibria (Theorem 3), in which agents vote informatively in the first round and vote for the alternative whose first-round vote exceeds a threshold. Such a class derives more natural equilibria, including the “informative + surprisingly popular” equilibrium, which resembles the “surprisingly popular” mechanism.

Furthermore, we show with mild assumptions that every one-round equilibrium gives a two-round equilibrium that is not more “complicated” (Theorem 4). Specifically, agents in the two-round voting mechanism can ignore the polling stage and play the one-round equilibrium in the second round. Therefore, the two-round voting mechanism has a natural equilibrium in every instance, including those where one-round voting fails to have a natural solution, and it can reach an informed majority decision whenever one-round voting can.

We also test the one-round voting and the two-round voting with synthetic experiments where we apply generative AI agents to simulate voters. When the signal distribution is biased, the AI agents powered by the reasoning model reach the informed majority decision more often in the two-round voting than in the one-round voting. We also observe Bayesian updates and vote switches (in the second round). Our experiments provide insights into future applications for AI-augmented or proxy votes in the future and serve as references for real-world experiments.

1.2 Related Work and Discussion

The study of the voting game with imperfectly informed agents originates from the famous Condorcet Jury Theorem [10]. Condorcet showed that the majority vote can reveal the “correct alternative” with probability converging to one as the number of voters increases, conditioned on (1) agents honestly reflecting their information in the votes (informative voting), and (2) the probability of an agent getting the correct information $p > 0.5$.

The seminal paper by Austen-Smith and Banks [2] is the first to study this problem in a game-theoretical setting. They show that informative voting is not compatible with the strategic behavior of voters. As a consequence, a large literature focuses on the existence of non-informative equilibrium and its effect on making a good decision. Wit [30], Feddersen and Pesendorfer [16], and Myerson [24] discovered the existence of a mixed strategy equilibrium that reveals the ground truth. Feddersen and Pesendorfer [15] propose a unique class of equilibria that reach the informed majority decision. Han et al. [21] reveal the equivalence between a strategy profile being an equilibrium and reaching a good decision. Nevertheless, none of them guarantees a simple or natural equilibrium.

Multiple works go “beyond” the one-round voting game, investigating and proposing more sophisticated mechanisms to aggregate information. Coughlan [11] shows that when agent preferences are restricted to similar, adding a communication phase before voting leads to a good decision. Their result does not extend to scenarios where agents have diverse preferences as in our setting, because they adopted a different solution concept (Bayesian-Nash equilibrium) and their techniques are very different from ours. Gerardi and Yariv [17] show that equilibrium outcomes in “deliberation+different voting rules” are the same yet diverge without deliberation. Goeree and Yariv [18] further show empirically that

these stable outcomes resemble those in one-round informative voting. Their paper possesses a different object and doesn’t examine whether outcomes lead to good decisions. Dekel and Piccione [13] show that sequential voting does not have new equilibria compared to one-round voting. This implies that sequential voting may not have a simple equilibrium. All papers are very different from ours in the model and the results. Chopra et al. [9] study the necessary knowledge for agents to strategize in the voting problems.

Our paper is also related to strategic information transmission between informed experts and an uninformed decision maker [1, 4, 5, 12, 23]. Even when experts strategically report their information, the decision maker can still collect enough information to make an optimal decision [3, 23]. Other directions include dependent signals [22, 26, 29], different competencies [7, 20, 25], more alternatives [19, 31], and empirical studies [6, 14].

2 Preliminaries

We adopt the information structure and the one-round voting game model from [21, 28].

Information Structure. n agents vote for two alternatives **A** and **R**. Two possible world states $\mathcal{W} = \{L, H\}$ reflect the overall tendency of the preferences, where **A** is more preferred in H , and **R** is more preferred in L . The world state cannot be observed directly by the agents. Let $P_H = \Pr[W = H] > 0$ and $P_L = \Pr[W = L] > 0$ be the common prior of the world states.

Each agent receives a signal in $\mathcal{S} = \{l, h\}$ related to the world state. We use m to denote a generic signal and S_i to denote the random variable representing the signal that agent i receives. We assume the signals are independent and have identical distributions conditioned on the world state. $P_{mk} = \Pr[S_i = m \mid W = k]$ is the probability that an agent receives signal m under world state k . The signal distributions $((P_{hH}, P_{lH}), (P_{hL}, P_{lL}))$ are common knowledge. The signals are positively correlated to the world states. We assume $P_{hH} > P_{hL}$ and $P_{lH} < P_{lL}$. We allow biased signals and DO NOT assume $P_{hH} > P_{lH}$ or $P_{hL} < P_{lL}$, which is a stronger assumption.

Utility and Types of Agents. Each agent i has a utility function on the true world state and the outcome of the vote. Formally, $v_i : \mathcal{W} \times \{\mathbf{A}, \mathbf{R}\} \rightarrow \{0, 1, 2, \dots, B\}$, where B is the integer upper bound. In our model, **A** is more welcomed in H , and **R** is more welcomed in L : $v_i(H, \mathbf{A}) \geq v_i(L, \mathbf{A})$, and $v_i(L, \mathbf{R}) \geq v_i(H, \mathbf{R})$.

Agents can be categorized into three types by their preferences. *Predetermined* agents always prefer the same alternative, and *contingent* agents have preferences depending on the world state. *Predetermined* agents can be further categorized into *friendly* and *unfriendly* agents based on the alternative they prefer. For an agent i , if i is a friendly agent, $v_i(H, \mathbf{A}) > v_i(L, \mathbf{A}) > v_i(L, \mathbf{R}) > v_i(H, \mathbf{R})$; if i is an unfriendly agent, $v_i(L, \mathbf{R}) > v_i(H, \mathbf{R}) > v_i(H, \mathbf{A}) > v_i(L, \mathbf{A})$; and if i is a contingent agent, $v_i(H, \mathbf{A}) > v_i(H, \mathbf{R})$ and $v_i(L, \mathbf{R}) > v_i(L, \mathbf{A})$.

Let F, U, C be the set of three types of agents, respectively, and α_F, α_U , and α_C be the approximated fraction of each type of agent. Given n agents, $n_F = \lfloor \alpha_F \cdot n \rfloor$ is the number of friendly agents, $n_U = \lfloor \alpha_U \cdot n \rfloor$ is the number of unfriendly agents, and $n_C = n - n_F - n_U$ is the number of contingent agents. α_F, α_U , and α_C are common knowledge and do not depend on n .

Informed Majority Decision. The goal of the voting is to output the *informed majority decision*, the alternative favored by the majority given the world state is known to all the agents. We assume that neither α_F nor α_U is larger than 0.5. Otherwise, friendly or unfriendly agents can dominate the voting outcome, and the alternative favored by the majority is independent of the world state. Therefore, in the rest of this paper, **A** is the informed majority decision under world state H , and **R** is the informed majority decision under world state L .

EXAMPLE 2 (INFORMATION STRUCTURE). Consider the COVID-19 policy-making scenario. Suppose $n = 20$ voters are deciding whether to accept (**A**) or reject (**R**) the policy. World state H represents a viral pandemic, and world state L represents the opposite. The common prior on the world state is $P_H = 0.6$, and $P_L = 0.4$. The signals are noisy information about the world state. The distribution is $P_{HH} = 0.8$, $P_{HL} = 0.2$, $P_{HL} = 0.6$, and $P_{LL} = 0.4$. An agent receives an h signal with probability 0.8 when the world state is H and probability 0.6 when the state is L .

Among 20 agents, there are 5 friendly agents, 6 unfriendly agents, and 9 contingent agents. Neither friendly nor unfriendly agents make up the majority. Therefore, the informed majority decision is **A** under world state H and **R** under world state L . We assume that agents of the same type share the same utility function (which may not be true in general), as shown in Table 2.

Table 2: Utility of agents in Example 2.

(Winner, State)	(A, H)	(A, L)	(R, H)	(R, L)
Friendly agent	7	4	1	3
Unfriendly agent	3	1	4	6
Contingent agent	5	2	3	4

Sequence of Environments. An environment $\mathcal{I}$ is a set of parameters of a voting game, which contains the agent number n , the world state prior distribution (P_L, P_H), the signal distributions (P_{hH}, P_{hL}) and (P_{hL}, P_{lL}), and the approximated fraction of each type of agents ($\alpha_F, \alpha_U, \alpha_C$).

In this paper, we study the problem asymptotically. Therefore, we define a sequence of environments in which the number of voters increases while most other parameters remain the same. Let $\{\mathcal{I}_n\}_{n=1}^\infty$ (or $\{\mathcal{I}_n\}$ for short) be a sequence of environments, where each $\mathcal{I}_n$ is an environment of n agents. The environments in a sequence share the same prior distributions, signal distributions, and approximated fractions of each type. We have no additional assumptions on the utility functions of agents.

2.1 One Round Voting

One-round voting. In the one-round voting game, every agent casts a vote for either **A** or **R**. The alternative getting more than a 0.5 fraction of votes becomes the winner.

Strategy in the one-round voting game. We use $\hat{\sigma}$ to denote the strategy of an agent in the one-round voting game. $\hat{\sigma}$ is a mapping from an agent's signal to a probability distribution on $\{\mathbf{A}, \mathbf{R}\}$ representing her action. A strategy is pure if the vote is deterministic. A strategy profile $\hat{\Sigma}$ is the vector of strategies of all agents. A strategy profile is pure if it contains only pure strategies and symmetric if

all agents with equal utility function play the same strategy. An agent votes *informatively* if he/she always votes for **A** when his/her signal is h and always votes for **R** when his/her signal is l .

Expected utility. Let $\lambda_k^A(\hat{\Sigma})$ ($\lambda_k^R(\hat{\Sigma})$, respectively) be the (ex-ante, before agents receiving their signals) probability that **A** (**R**, respectively) becomes the winner in the one round voting when the world state is k . The expected utility of an agent i exclusively depends on $\lambda_k^A(\hat{\Sigma})$ and $\lambda_k^R(\hat{\Sigma})$:

$$u_i(\hat{\Sigma}) = P_L(\lambda_L^A(\hat{\Sigma}) \cdot v_i(L, \mathbf{A}) + \lambda_L^R(\hat{\Sigma}) \cdot v_i(L, \mathbf{R})) \\ + P_H(\lambda_H^A(\hat{\Sigma}) \cdot v_i(H, \mathbf{A}) + \lambda_H^R(\hat{\Sigma}) \cdot v_i(H, \mathbf{R})).$$

Sequence of Strategy Profiles. We define a sequence of strategy profiles $\{\hat{\Sigma}_n\}_{n=1}^\infty$ on an environment sequence $\{\mathcal{I}_n\}$, where for each n , $\hat{\Sigma}_n$ is a strategy profile of a one-round voting game with parameters in $\mathcal{I}_n$. We do not have additional assumptions about the agents. Therefore, for different n , the strategies of agents can be drastically different without further specification.

ε -strong Bayes-Nash Equilibrium (BNE). Due to the nature of the majority rule, deviation from a single agent is unlikely change the outcome. Therefore, we study the group strategic behavior of agents and adopt the solution concept of (approximated) strong Bayes-Nash equilibrium. A strategy profile $\hat{\Sigma} = (\hat{\sigma}_1, \hat{\sigma}_2, \dots, \hat{\sigma}_n)$ is an ε -strong Bayes-Nash Equilibrium if there does not exist a subset of agents D (the *deviating group*) and a strategy profile $\hat{\Sigma}' = (\hat{\sigma}'_1, \hat{\sigma}'_2, \dots, \hat{\sigma}'_n)$ (the *deviating strategy profile*) such that (1) $\hat{\sigma}_i = \hat{\sigma}'_i$ for all $i \notin D$; (2) $u_i(\hat{\Sigma}') \geq u_i(\hat{\Sigma})$ for all $i \in D$; and (3) there exists $i \in D$ such that $u_i(\hat{\Sigma}') > u_i(\hat{\Sigma}) + \varepsilon$. When $\varepsilon = 0$, the equilibrium is a strong Bayes-Nash Equilibrium where no group of deviators can strictly increase their utilities.

3 Equilibria in One-Round Voting

We show scenarios where natural strategy profiles fail to reach high fidelity (and thus are not approximate strong BNEs according to the if-and-only-if characterization in [21]) and how complex the existing equilibria are. Firstly, in games with only contingent agents with identical utility functions, NO pure symmetric equilibria exist in the one-round voting game in ANY environment with biased signals.

PROPOSITION 1. Suppose $\alpha_C = 1$. For any sequence of environments $\{\mathcal{I}_n\}$ such that $P_{hL} > 0.5$ or $P_{lH} > 0.5$ and that for each n , all agents share the same utility function v , no sequence pure symmetric strategy profiles $\{\hat{\Sigma}_n\}$ of the one-round voting game satisfies that $\hat{\Sigma}_n$ is a ε -strong BNE for every n with $\varepsilon \rightarrow 0$.

Note that our definition of symmetry (agents with the same utility function play the same strategy) is very broad. Therefore, Proposition 1 excludes a large number of strategy profiles from being equilibria. The proof of Proposition 1 is in the full version.

In the environments where pure symmetric strategies fail to be equilibria, reaching the informed majority decision requires agents to artificially “shift” the probabilities in the strategy. For example, consider a biased setting where the common prior is $P_H = P_L = 0.5$, and the signal distribution is $P_{hH} = 0.9, P_{lH} = 0.1, P_{hL} = 0.7$, and $P_{lL} = 0.3$. An equilibrium under this setting is a symmetric mixed profile where agents with signal l always vote for **R**, and agents

with h vote for **A** with probability 0.6. In this strategy profile, the informed majority decision receives more than half of the vote in expectation. Alternatively, agents can play a pure asymmetric profile where 40% of agents always vote for **R**.

These “successful” strategy profiles, however, are often too complex for agents to agree on. For the mixed strategy profile, even if agents are clever enough to be aware of a probability shift, different agents may choose different probability shifts, which may still lead to the undesirable outcome. For the pure asymmetric profile, without an intrinsic incentive, it is difficult and unclear for identical agents to coordinate who casts the “opposite vote” **R** when receiving signal h .

4 Two-Round Voting and Equilibria

Two-round voting and one-round voting share the same environment, including the number of agents, distributions, the fraction of each type, and utilities. Their difference lies in the game form and the strategies. Agents play a two-round anonymous voting game where the winner is completely determined by the second round. The five steps are shown in Algorithm 1.

ALGORITHM 1: Two-round voting mechanism

- 1: All the agents receive their signals.
 - 2: **First round** (Polling stage): agents cast the first round vote for **A** or **R**.
 - 3: Agents observe the number of votes for **A** in the first round denoted by $x \in \{0, 1, \dots, n\}$.
 - 4: **Second Round** (Voting stage): agents cast the second round vote for **A** or **R**.
 - 5: The alternative that gets more than a 0.5 fraction of votes in the second round becomes the winner.
-

Strategy. A strategy in a two-round voting game σ_i is a pair (σ_i^1, σ_i^2) . The first-round strategy σ_i^1 is a mapping from agent i 's signal to a distribution on $\{\mathbf{A}, \mathbf{R}\}$ denoting her action in the first round. The second-round strategy σ_i^2 is a mapping from agent i 's signal AND the result of the first-round vote x to a distribution on $\{\mathbf{A}, \mathbf{R}\}$ denoting her action in the second round. A strategy profile Σ is the vector of strategies of all agents. The definition of pure and symmetric strategy profile follows that in the one-round voting.

We define a sequence of two-round voting strategy profiles $\{\Sigma_n\}_{n=1}^\infty$ on an environment sequence $\{I_n\}$. For each n , Σ_n is a profile of a two-round voting game with parameters in I_n .

Expected Utility and Fidelity. Given a strategy profile Σ , let $\lambda_k^{\mathbf{A}}(\Sigma)$ ($\lambda_k^{\mathbf{R}}(\Sigma)$, respectively) be the—ex-ante, before agents receiving their signals—probability that **A** (**R**, respectively) becomes the winner when the world state is k . The expected utility in the two-round voting is in the same form as that in the one-round voting.

$$u_i(\Sigma) = P_L(\lambda_L^{\mathbf{A}}(\Sigma) \cdot v_i(L, \mathbf{A}) + \lambda_L^{\mathbf{R}}(\Sigma) \cdot v_i(L, \mathbf{R})) \\ + P_H(\lambda_H^{\mathbf{A}}(\Sigma) \cdot v_i(H, \mathbf{A}) + \lambda_H^{\mathbf{R}}(\Sigma) \cdot v_i(H, \mathbf{R})).$$

Fidelity is the likelihood that the informed majority decision is reached. In the two-round voting game, the fidelity of a strategy profile Σ is as follows.

$$A(\Sigma) = P_L \cdot \lambda_L^{\mathbf{R}}(\Sigma) + P_H \cdot \lambda_H^{\mathbf{A}}(\Sigma).$$

An ε -strong BNE in the two-round voting has the same definition as that in the one-round voting, except that the strategies and expected utilities are those in the two-round voting.

Remarks. The game of two-round voting is not exactly a standard *Bayesian game*, as the strategy of an agent also depends on the first round outcome. Our definitions of strategies, utilities, and equilibria are straightforward extensions of the standard Bayesian games.

Another natural formulation for the two-round voting is an *extensive-form game*. We believe this is an interesting direction for future work. However, standard equilibrium solution concepts such as *subgame perfect Nash equilibrium* and *perfect Bayesian equilibrium* fail to capture the feature of group deviations. The first step in this direction is to propose a reasonable equilibrium concept that extends the strong Bayes-Nash equilibrium to extensive-form games.

4.1 Equilibria in Two Round Voting

We show that the two-round voting mechanism indeed has natural equilibria leading to the informed majority decision. First, we show that every ε -strong BNE in the two-round voting mechanisms with ε converges to 0 reaches the informed majority decision with high probability.

THEOREM 1 (All ε -BNE are good). *Let $\{\Sigma_n\}$ be a sequence of profiles in the two-round voting game. If for any n , Σ_n is an ε -strong BNE with $\varepsilon = o(1)$, then $\lim_{n \rightarrow \infty} A(\Sigma_n) = 1$.*

Moreover, we show the existence of a natural equilibrium in the two-round mechanism. While the classical informative voting or sincere voting may not be an equilibrium in the one-round voting game, their combination becomes an equilibrium.

“Informative + Sincere” voting. All the friendly agents always vote for **A** in both rounds, and all the unfriendly agents always vote for **R** in both rounds. All the contingent agents vote informatively in the first round and vote sincerely based on their observation of the first round outcome, as shown in Definition 1.

Definition 1 (Sincere Voting in the second round.). The agent compares his/her expected utility between the event that **A/R** becomes the winner conditioned on his/her private signal m , the first-round outcome x . He/she vote for **A** if and only if $u_i(\mathbf{A} | x, m) \geq u_i(\mathbf{R} | x, m)$.

$$u_i(\mathbf{A} | x, m) = \Pr[W = L | x, m] \cdot v_i(L, \mathbf{A}) + \Pr[W = H | x, m] \cdot v_i(H, \mathbf{A}). \\ u_i(\mathbf{R} | x, m) = \Pr[W = L | x, m] \cdot v_i(L, \mathbf{R}) + \Pr[W = H | x, m] \cdot v_i(H, \mathbf{R}).$$

THEOREM 2 (“Informative + sincere” equilibrium). *For any environment sequence $\{I_n\}$, the sequence of the “information + sincere” strategy profiles $\{\Sigma_n^\dagger\}$ satisfies that (1) $\lim_{n \rightarrow \infty} A(\Sigma_n^\dagger) = 1$, and (2) for every n , $\Sigma_n^\dagger$ is a ε -strong BNE where ε converges to 0 as $n \rightarrow \infty$.*

The rationale behind the “informative + sincere” strategy is straightforward—by observing the outcome of the first-round informative voting, every contingent agent knows the private information of all contingent agents anonymously, which informs them of the correct world state with high probability. Therefore, in the second round, they are nearly certain of their preferred alternative and vote accordingly.

While agents may not necessarily perform Bayesian updates in the votes, we show that the intuition of “sharing information and getting updated” derives a large class of “informative + threshold” equilibria (including the “informative + sincere” equilibrium). Therefore, as agents still follow this “informative + threshold” pattern, the informed majority can be reached.

In an “informative + threshold” strategy profile, all the predetermined agents vote for their preferred alternative in both rounds; every contingent agent i votes informatively in the first round and votes for **A** in the second round if and only if the first round outcome x exceeds some threshold x_i^* . For the “informative + sincere” voting, the threshold x^* is determined by the constraint $u_i(\mathbf{A} \mid x, m) \geq u_i(\mathbf{R} \mid x, m)$, as agents apply the Bayes Theorem to update their belief. (see Appendix B.) Additionally, “informative + threshold” class also includes more “natural” strategy profiles, such as the “informative + surprisingly popular” strategy profile.

EXAMPLE 3. *The “informative + surprisingly popular” strategy shares the similar idea with the “surprisingly popular” mechanism [27]. The threshold of every agent i is the prior expectation on the first-round outcome x , which is $x^* = n_F + n_C \cdot (P_H \cdot P_{hH} + P_L \cdot P_{hL})$. This comes from that the n_F friendly agents always vote for **A**, and the expected share of n_C contingent agents voting for **A** is P_{hH} under world state H and P_{hL} under world state L . When the number of votes for **A** exceeds the expectation, **A** becomes the “surprisingly popular” alternative and is elected the winner in the second round. Otherwise, **R** becomes the “surprisingly popular” alternative.*

The “informative + threshold” strategy profiles share the merit that the “informative + sincere” equilibrium has. Every agent votes deterministically. When agents with equal utility functions adopt the same threshold, they are also symmetric. Most importantly, every such strategy profile in which all thresholds lie between the expected first-round outcome conditioned on two world states is an equilibrium that leads to the informed majority decision.

Definition 2. We say an sequence of “informative + threshold” strategies is constantly separable if there exists a $n_0 > 0$ and a constant $\delta > 0$ such that for every $n \geq n_0$ and every agent i , $n_F + n_C \cdot P_{hL} + \delta \cdot n \leq x_i^* \leq n_F + n_C \cdot P_{hH} - \delta \cdot n$. (See Figure 1.)

THEOREM 3 (“Informative + threshold” equilibrium). *Under any environment sequence $\{I_n\}$, every constantly separable sequence of profiles $\{\Sigma_n^*\}$ satisfies that (1) $\lim_{n \rightarrow \infty} A(\Sigma_n^*) = 1$, and (2) for every n , Σ_n^* is a ϵ -strong BNE where ϵ converges to 0 as n goes to ∞ .*

Remark on Theorem 1. Theorem 1 states that any $o(1)$ -strong BNE leads to nearly optimal fidelity. Unlike the one-round voting, the other direction—optimal fidelity implying equilibria—does not hold in the two-round voting game, even when predetermined agents only play their dominant second-round strategies. For example, friendly and unfriendly agents voting informatively in the first round can help contingent agents infer the world state and lead to an informed majority decision. However, this is not an equilibrium, as friendly agents have incentives to always vote for **A** in the first round and deceive contingent agents that the world state is H . Likewise, unfriendly agents have incentives to vote for **R**.

4.2 Proof Sketch for Theorem 1

Suppose $\{\Sigma_n\}$ is a sequence of strategy profiles whose fidelity does not converge to 1. We show that there exists a different sequence of profiles $\{\Sigma'_n\}$ such that agents have incentives to deviate from Σ_n to Σ'_n for infinitely many n . We first give the proof when $\{\Sigma_n\}$ is a *regular* profile sequence, where all friendly agents always vote for **A** and all unfriendly agents always vote for **R** in the *second* round. This helps us characterize the main idea while reducing complications. Then we extend the proof to non-regular profiles. The proof complementing details is the full version.

Proof for regular profiles. We take an infinite set $\mathcal{N} \subseteq \mathbb{N}_+$ and a constant $\delta > 0$ such that for all $n \in \mathcal{N}$, the fidelity of Σ_n does not exceed $1 - \delta$. Let the deviating group D be all the contingent agents. If Σ'_n increases the fidelity, contingent agents get higher expected utility and are incentivized to deviate. Therefore, we aim to construct Σ'_n so that contingent agents reveal the world state with high probability in the first round and vote for the (likely) informed majority decision in the second round. Given that predetermined agents vote for their preferred alternative in the second round, the informed majority decision becomes the winner.

Contingent agents can (collectively, with high probability) reveal the world states because signal distributions are different in different states of the world. The expected number of h signals from contingent agents is $P_{hH} \cdot n_C$ in world state H and $P_{hL} \cdot n_C$ in world state L . If all the contingent agents vote informatively in the first round, by the Hoeffding inequality, the number of votes will concentrate to the expectation with high probability. The expected votes for **A** is higher in world state H than in world state L , as $P_{hH} > P_{hL}$. Therefore, contingent agents can play a “threshold” strategy (between $P_{hH} \cdot n_C$ and $P_{hL} \cdot n_C$) in the second round: vote for **A** together if the first round result x exceeds the threshold and vote for **R** together otherwise.

However, such a strategy still must overcome one difficulty: strategies of predetermined agents in the first round are not guaranteed to be “regular” and may offset the difference in the votes of contingent agents. For example, predetermined agents can vote for **R** more often in world state H than in world state L in Σ_n so that x has a smaller expectation in world state H than in world state L . However, contingent agents can “reverse” their votes and observations accordingly: they vote opposite to their signal in the first round and vote for **A** if x is below a threshold. In this way, they can still distinguish the different world states.

Now we explicitly present the strategy profile Σ'_n . Let $\varphi_H(\Sigma'_n)$ and $\varphi_L(\Sigma'_n)$ be the expectation of the first-round outcome x of Σ'_n conditioned on the world state H and L respectively.

1. All predetermined agents (they do not deviate) play the same strategy as in Σ_n . This includes that they play their dominant second-round strategies.

2. If more votes for **A** from predetermined agents are expected in world state H than in L in the first round, all contingent agents vote informatively in the first round. For the second round, they vote for **A** if $x \geq \frac{\varphi_H(\Sigma'_n) + \varphi_L(\Sigma'_n)}{2}$ and vote for **R** otherwise.

3. Otherwise, contingent agents “reverse” the strategy. In the first round, they vote for **A** with l and **R** with h . In the second round, they vote for **A** if $x < \frac{\varphi_H(\Sigma'_n) + \varphi_L(\Sigma'_n)}{2}$ and **R** otherwise.

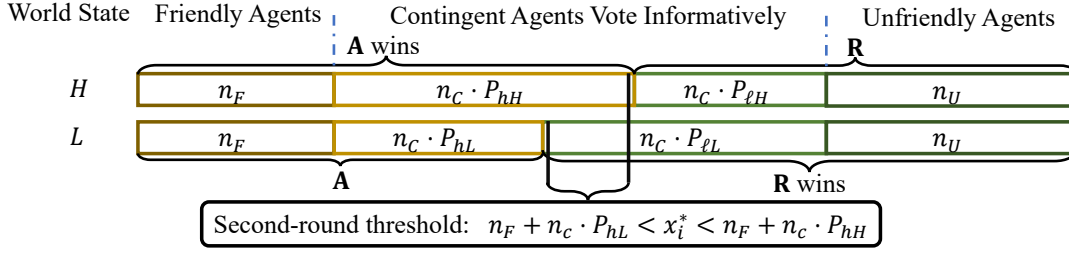


Figure 1: Illustration of the range of the threshold for an “informative + threshold” equilibrium.

In this way, the fidelity $A(\Sigma'_n)$ converges to 1 as n goes to infinity, contingent agents have incentives to deviate, and Σ_n is NOT an ε -strong BNE for some constant ε .

Proof Sketch for non-regular profiles. We adopt the same idea to construct a strategy profile where the deviating group forces the state to be revealed by the first-round vote and votes for the preferred agent in the second round. The difficulty for the non-regular case is that a deviating group with only contingent agents may not work, as there is no assumption on the behavior of predetermined agents in the second round. (For example, if $\alpha_C < 0.5$, and all the predetermined agents always vote for A, then the winner will always be A.) Fortunately, the following lemma shows that there will always be a type of predetermined agents who have incentives to deviate together to reach the informed majority decision.

LEMMA 1. *Let $\{\Sigma_n\}$ be a sequence of profiles such that $A(\Sigma_n)$ does not converge to 1. Then there exists infinitely many n such that for each n , at least one of the following holds for all sequences of profiles $\{\Sigma'_n\}$ with $\lim_{n \rightarrow \infty} A(\Sigma'_n) = 1$: (1) For every friendly agent i_1 , $u_{i_1}(\Sigma'_n) - u_{i_1}(\Sigma_n) \geq 0$. (2) For every unfriendly agent i_2 , $u_{i_2}(\Sigma'_n) - u_{i_2}(\Sigma_n) \geq 0$. Moreover, for a fixed n and Σ_n , either (1) holds for all Σ'_n or (2) holds for all Σ'_n .*

In this way, we construct two deviating strategies. In the first strategy profile Σ_F , all the friendly and contingent agents play an “informative + threshold” strategy. The second strategy profile Σ_U is defined similarly. With similar reasoning to the regular case, both Σ_F and Σ_U have fidelity converging to 1. Applying Lemma 1, there is a type of pre-determined agents that get expected utility increase via deviation in both Σ_F and Σ_U . Contingent agents will also get a constant increase in expected utility due to the increase in fidelity. Therefore, agents have incentives to deviate to either Σ_F or Σ_U , and Σ_n is NOT an ε -strong BNE with constant ε .

4.3 Proof Sketch for Theorem 2 and 3

As the “informative + sincere” strategy profile is also an “informative + threshold” strategy profile, it suffices to display the proof for Theorem 3 and shows that the sincere threshold satisfies the constantly separable constraint. The proof of Theorem 3 consists of two parts: Lemma 2 shows that the fidelity converges to 1, and Lemma 3 shows that the profiles are $o(1)$ -strong BNE. The full proof is in the full version of this paper.

LEMMA 2. *For any sequence of instance and any sequence of “informative + threshold” strategy profiles Σ , if there exists a constant*

$\delta > 0$ such that for all n and all agent i , $n_F + n_C \cdot P_{hL} + \delta \cdot n \leq x_i^n \leq n_F + n_C \cdot P_{hH} - \delta \cdot n$, then $A(\Sigma)$ converges to 1.

Lemma 2 follows similar reasoning in Theorem 1 of showing the deviating strategy profile have fidelity converging to 1. Firstly, the threshold x^* is guaranteed to be between the expectation of the first-round vote conditioned on the world state being H and being L . Therefore, with probability converging to 1, the first-round outcome is larger than all the thresholds under world state H and lower than all the thresholds under world state L . Therefore, the contingent agents know the world state and vote for the informed majority decision with high probability.

LEMMA 3. Σ_n^* is an ε -strong BNE with $\varepsilon = o(1)$.

The proof of Lemma 3 follows a similar idea to the proof of [28, Theorem 3.3]. For arbitrary n , and a strategy profile Σ_n , let $\varepsilon = 2B(B+2)(1-A(\Sigma_n))$, where B is the upper bound of the utility. Then, for any deviating strategy profile Σ'_n , we show that the deviation will not succeed. Therefore, Σ_n is an ε -strong BNE. Since $A(\Sigma_n^*)$ converges to 1, ε converges to 0.

LEMMA 4 (INFORMAL). *For any strategy profile Σ , a deviating group D that have incentives to deviate from Σ to another strategy profile Σ' , under the approximation parameter $\varepsilon = 2B(B+2)(1-A(\Sigma))$ for ε -strong BNE, contains either only friendly agents or only unfriendly agents.*

The proof of Lemma 4 consists of two cases. If $1 - A(\Sigma') < 2 \cdot (1 - A(\Sigma))$, the fidelities of both profiles are close to 1, and the expected utilities of agents will have no significant difference in two profiles. Therefore, no agents can get more than ε expected utility gain by deviation. Otherwise, we show that the deviating group contains only one type of predetermined agents. Firstly, contingent agents, whose expected utilities align with the fidelity, are excluded from the deviating group D . Then, friendly agents and unfriendly agents cannot be in D simultaneously, as an expected utility gain of more than ε on one side implies an expected utility decrease on the other side. The formal statement of Lemma 4 is in the full version.

Then we show that D contains only friendly agents or only unfriendly agents cannot succeed. Without loss of generality, suppose D contains only friendly agents. If friendly agents deviate in the first round, the result of the first-round voting x will be less likely to reach the threshold, and contingent agents are more likely to vote for R. If friendly agents deviate in the second round, there will be strictly fewer expected votes for A, and A will be less likely to be the winner. Therefore, no deviation succeeds, and Σ_n^* is a ε -strong BNE with ε converges to 0.

Finally, we show that the “informative+sincere” threshold is between the two expectation. It shares the same idea with Lemma 2. Recall that for any agent i , the threshold x_i^* is the boundary for $u_i(A | x, m) \geq u_i(R | x, m)$. When x is close to the upper bound (expected utility when the world state is H), $\Pr[H | x, m]$ is close to 1, and $u_i(A | x, m) \geq u_i(R | x, m)$ always hold (as contingent agents prefer A in world state H); and when x is close to the lower bound (expected utility when the world state is L), $\Pr[H | x, m]$ is close to 0, and $u_i(A | x, m) < u_i(R | x, m)$ always hold. Therefore, the threshold must be in between.

4.4 Two-Round Voting is More Powerful

In this section, we show that in any sequence of environments where there exists a one-round equilibrium (where predetermined agents play their dominant strategy), there always exists a two-round equilibrium that will not be more “complicated” than the one-round equilibrium. Precisely, in this two-round equilibrium, agents disregard the first-round voting and play exactly as in the one-round equilibrium in the second round. By not exploiting the information revelation in the first round, the two-round voting was reduced to a one-round voting game. (A one-round voting strategy profile is *regular* if all predetermined agents play their dominant strategies.)

Therefore, we claim that two-round voting is more powerful than one-round voting, as it can reach an informed majority decision whenever one-round voting can reach it, and it provides a natural equilibrium in every environment including those where one-round voting fails to have a natural solution. The proof resembles that of Theorem 3.

THEOREM 4. *Under the same sequence of environments, for any regular one-round voting strategy profile sequence being $o(1)$ -strong BNE, any two-round voting profile sequence, where agents play exactly this one-round strategy in the second round, is also an $o(1)$ -strong BNE.*

5 Experiments

We conducted simulated voting experiments with generative AI agents to compare one-round and two-round voting. We inject AI bots with different preferences and information drawn from given distributions, then run both voting under various environments.

We adopt two different signal distributions. the unbiased distribution $P_{hH} = 0.8, P_{hL} = 0.2$ and the biased distribution $P_{hH} = 0.8, P_{hL} = 0.6$. The selection of the distribution is based on a wider range test of different distributions to capture different scenarios anticipated by our theoretical findings on the performance of one-round versus two-round voting. Under the unbiased distribution, both mechanisms admit natural equilibria that achieve the informed majority decision. In contrast, the biased distribution admits no pure-symmetric equilibria for one-round voting, whereas “informative + sincere” remains viable. The signal that an agent receives can be precise (with exact numbers) and vague (with word descriptions).

Under each implementation of world state and private signals, we run a one-round voting and a two-round voting. In one-round voting, each agent was asked only once for its vote. In two-round voting, each agent voted twice: we first collected every agent’s first-round vote simultaneously, then shared the first-round outcome

with all agents (including their own first-round response as a history message) when asking for the second-round vote. More details of the experimental implementation is in Appendix C.

5.1 Experimental Results

Table 3: The likelihood of the informed majority decision is reached. DS is the abbreviation for Deepseek.

Model	Distribution	Unbiased (0.8, 0.2)		Biased (0.8, 0.6)	
	Information	Precise	Vague	Precise	Vague
DS R1	One-round	1	1	0.56	0.52
	Two-round	1	1	0.88	0.78
GPT-4o	One-round	1	1	0.5	0.52
	Two-round	1	1	0.48	0.5

As shown in Table 3, when signals are unbiased, both voting mechanisms always reach the informed majority decision. However, on the biased signal distribution, Deepseek R1 agents show a marked accuracy gap: one-round voting frequently misses the informed majority decision, whereas two-round voting retains good performance under both world states.

We also examine the behaviors of the AI agents from their reasoning and analysis, and have some interesting observations. Firstly, almost every agent performs a Bayesian update to infer the likelihood of the world state based on the signal distribution and the private information it obtains. Even with vague information, agents tend to guess a concrete value for each description and perform the Bayesian update. Secondly, we observe some agents cast different second-round votes. They usually find considerable votes in the first round that contradict their beliefs, which drives them to switch. On the other hand, there are also agents who believe more and stick to their one belief. An example of the answers is in Appendix C.

Our experiments illustrate a scenario where two-round voting is more likely to lead to an informed majority decision than one-round voting played by generative AI agents. These findings can guide future applications of generative AI in voting systems, such as AI-augmented or AI-proxy voting. Moreover, our experiments serve as a practical reference for designing real-world experiments that compare voting rules under strategic behavior.

6 Conclusion

We study the voting problem with two alternatives where voters’ preferences depend on a not-directly-observable state variable and propose the two-round voting mechanism that reaches the informed majority decision via natural equilibria. This sharply contrasts with one-round voting mechanisms that often fail to provide both simplicity and effective equilibria simultaneously.

A future direction is to formulate the two-round voting game as an extensive-form game. We need to first find a reasonable equilibrium concept that extends the strong Bayes-Nash equilibrium to the extensive-form game. It would be interesting to see if the corresponding equilibria lead to good election outcomes. Another possible direction is to conduct human-subject experiments comparing one-round and two-round voting mechanisms, as a further step of our generative AI experiments.

References

- [1] David Austen-Smith. 1993. Interested experts and policy advice: Multiple referrals under open rule. *Games and Economic Behavior* 5, 1 (1993), 3–43.
- [2] David Austen-Smith and Jeffrey S. Banks. 1996. Information Aggregation, Rationality, and the Condorcet Jury Theorem. *The American Political Science Review* 90, 1 (1996), 34–45.
- [3] Eyal Baharad, Jacob Goldberger, Moshe Koppel, and Shmuel Nitzan. 2012. Beyond Condorcet: Optimal aggregation rules using voting records. *Theory and Decision* 72, 1 (2012), 113–130.
- [4] Marco Battaglini. 2002. Multiple referrals and multidimensional cheap talk. *Econometrica* 70, 4 (2002), 1379–1401.
- [5] Marco Battaglini. 2004. Policy advice with imperfectly informed experts. *The BE Journal of Theoretical Economics* 4, 1 (2004), 0000102202153459631100.
- [6] Marco Battaglini, Rebecca B. Morton, and Thomas R. Palfrey. 2010. The Swing Voter's Curse in the Laboratory. *The Review of Economic Studies* 77, 1 (01 2010), 61–89.
- [7] Ruth Ben-Yashar and Mor Zahavi. 2011. The Condorcet jury theorem and extension of the franchise with rationally ignorant voters. *Public Choice* 148, 3 (2011), 435–443.
- [8] Duncan Black. 1948. On the rationale of group decision-making. *Journal of Political Economy* 56, 1 (1948), 23–34.
- [9] Samir Chopra, Eric Pacuit, and Rohit Parikh. 2004. Knowledge-Theoretic Properties of Strategic Voting. In *Logics in Artificial Intelligence*, José Júlio Alferes and João Leite (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 18–30.
- [10] Marquis de Condorcet. 1785. *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*. Paris: L'Imprimerie Royale.
- [11] Peter J. Coughlan. 2000. In Defense of Unanimous Jury Verdicts: Mistrials, Communication, and Strategic Voting. *The American Political Science Review* 94, 2 (2000), 375–393.
- [12] Vincent P Crawford and Joel Sobel. 1982. Strategic information transmission. *Econometrica: Journal of the Econometric Society* (1982), 1431–1451.
- [13] Eddie Dekel and Michele Piccione. 2000. Sequential Voting Procedures in Symmetric Binary Elections. *Journal of Political Economy* 108 (2000), 34–55. Issue 1.
- [14] Ignacio Esponda and Emanuel Vespa. 2014. Hypothetical Thinking and Information Extraction in the Laboratory. *American Economic Journal: Microeconomics* 6, 4 (November 2014), 180–202.
- [15] Timothy Feddersen and Wolfgang Pesendorfer. 1997. Voting Behavior and Information Aggregation in Elections With Private Information. *Econometrica* 65, 5 (1997), 1029–1058.
- [16] Timothy Feddersen and Wolfgang Pesendorfer. 1998. Convicting the Innocent: The Inferiority of Unanimous Jury Verdicts under Strategic Voting. *American Political Science Review* 92, 1 (1998), 23–35.
- [17] Dino Gerardi and Leeat Yariv. 2007. Deliberative voting. *Journal of Economic Theory* 134, 1 (2007), 317–338.
- [18] Jacob K. Goeree and Leeat Yariv. 2011. An Experimental Study of Collective Deliberation. *Econometrica* 79, 3 (2011), 893–921.
- [19] Johanna M.M. Goertz and François Maniquet. 2014. Condorcet Jury Theorem: An example in which informative voting is rational but leads to inefficient information aggregation. *Economics Letters* 125, 1 (2014), 25–28.
- [20] Mark Gradstein and Shmuel Nitzan. 1987. Organizational decision-making quality and the severity of the free-riding problem. *Economics Letters* 23, 4 (1987), 335–339.
- [21] Qishen Han, Grant Schoenebeck, Biaoshuai Tao, and Lirong Xia. 2023. The Wisdom of Strategic Voting. In *Proceedings of the 24th ACM Conference on Economics and Computation* (London, United Kingdom). Association for Computing Machinery, New York, NY, USA, 885–905.
- [22] Serguei Kaniovski. 2010. Aggregation of correlated votes and Condorcet's Jury Theorem. *Theory and Decision* 69, 3 (2010), 453–468.
- [23] John Morgan and Phillip C. Stocken. 2008. Information Aggregation in Polls. *American Economic Review* 98, 3 (June 2008), 864–96. doi:10.1257/aer.98.3.864
- [24] Roger B. Myerson. 1998. Extended Poisson Games and the Condorcet Jury Theorem. *Games and Economic Behavior* 25, 1 (1998), 111–131.
- [25] Shmuel Nitzan and Jacob Paroush. 1980. Investment in Human Capital and Social Self Protection under Uncertainty. *International Economic Review* 21, 3 (1980), 547–557.
- [26] Shmuel Nitzan and Jacob Paroush. 1984. The significance of independent decisions in uncertain dichotomous choice situations. *Theory and Decision* 17, 1 (1984), 47–60.
- [27] Dražen Prelec, H Sebastian Seung, and John McCoy. 2017. A solution to the single-question crowd wisdom problem. *Nature* 541, 7638 (2017), 532–535.
- [28] Grant Schoenebeck and Biaoshuai Tao. 2021. Wisdom of the Crowd Voting: Truthful Aggregation of Voter Information and Preferences. In *Advances in Neural Information Processing Systems*, Vol. 34. Curran Associates, Inc., 1872–1883.
- [29] Lloyd Shapley and Bernard Grofman. 1984. Optimizing group judgmental accuracy in the presence of interdependencies. *Public Choice* 43, 3 (1984), 329–343.
- [30] Jürgen Wit. 1998. Rational Choice and the Condorcet Jury Theorem. *Games and Economic Behavior* 22, 2 (1998), 364–376.
- [31] H. Peyton Young. 1988. Condorcet's Theory of Voting. *American Political Science Review* 82 (1988), 1231–1244.

All the technical proofs can be found in the full version of this paper at <https://arxiv.org/abs/2505.10377>.

A Concrete Calculation for Example 1

Here we give a slight more concrete calculation on Example 1 on a fixed number of agents. Suppose there are $n = 5$ agents. Consider agent i 's expected utility. Suppose i 's signal is High, and let u_i and u'_i be his/her expected utility on voting for Accept (align with High) and Reject (align with Low), respectively. According to the analysis, i 's vote changes the outcome if and only if exactly two of the other four agents receive signal High and vote for Accept. Therefore, the expected utility of i becomes the likelihood that the winning decision aligns with the real-world state, conditioned on this pivotal case (where the histogram of the signals is revealed). The difference between expected utility then becomes

$$u_i - u'_i = \Pr[\text{World state is High} \mid 3 \text{ agents receives High and 2 agents receive Low}] \\ - \Pr[\text{World state is Low} \mid 3 \text{ agents receives High and 2 agents receive Low}].$$

In abbreviation, we rewrite this as $u_i - u'_i = \Pr[W = H \mid 3H2L] - \Pr[W = L \mid 3H2L]$. By the Bayesian Theorem,

$$\begin{aligned} \Pr[W = H \mid 3H2L] &= \frac{\Pr[3H2L \mid H] \Pr[H]}{\Pr[3H2L \mid H] \Pr[H] + \Pr[3H2L \mid L] \Pr[L]} \\ &= \frac{0.5 \times 0.9^3 \times 0.1^2}{0.5 \times 0.9^3 \times 0.1^2 + 0.5 \times 0.6^3 \times 0.4^2} \\ &\approx 0.174. \\ \Pr[W = L \mid 3H2L] &\approx 0.826. \end{aligned}$$

Therefore, i has incentives to misreport his/her signal, and informative voting is not a Nash equilibrium in this scenario.

B Explicit form of “Informative + Sincere” Voting

In the “Informative + Sincere” strategy profile, a contingent agent i vote for **A** in the second round if and only if $x \geq x_i^*$, where

$$x_i^* = n_F + \frac{\log \frac{P_L \cdot (v_i(L,R) - v_i(L,A))}{P_H \cdot (v_i(H,A) - v_i(H,R))} + n_C \cdot \log \frac{P_{EL}}{P_{EH}}}{\log \frac{P_{HH}P_{EL}}{P_{HL}P_{HL}}}.$$

EXAMPLE 4. We calculate the “informative + sincere” strategy under the setting in Example 2. In this environment, $n_F = 5$, $n_C = 9$, $P_H = 0.6$, $P_{HH} = 0.8$, and $P_{HL} = 0.6$. Therefore, $x_i^* = 5 + \frac{\log \frac{2}{3} + 9 \times \log 2}{\log \frac{8}{3}} \approx 10.95$. Therefore, the contingent voters who vote sincerely in the second round will vote for **A** if and only if there are at least 11 votes for **A** in the first round. $\square$

C More Details on Experiments

C.1 Setting and Configuration

Our experiments employed two generative AI models—OpenAI GPT-4o and Deepseek R1. In each voting scenario, all voters were drawn from the same model. The experiment is run on a Lenovo Thinkbook 15 G3 AGL with AMD Ryzen 5 5600U with Radeon Graphics 2.30 GHz. The program is written in Python 3.9. We adopt online APIs to create generative AI agents. The API for Deepseek R1 is provided by Together AI, and the API for GPT-4o is provided by OpenAI’s official API platform.

For each vote, we set $n = 40$ agents, including 5 friendly agents, 3 unfriendly agents, and 32 contingent agents. The prior is fixed at $P_H = 0.5$. We run experiments on two different signal distributions, the unbiased distribution $P_{HH} = 0.8$, $P_{HL} = 0.2$ and the biased distribution $P_{HH} = 0.8$, $P_{HL} = 0.6$. We also test a wider range of signal distributions with fewer samples and select the distributions that capture different scenarios anticipated by our theoretical findings on the performance of one-round versus two-round voting. Under the unbiased distribution, both mechanisms admit natural equilibria that achieve the informed majority decision, so we expect strong performance from both. In contrast, the biased distribution, as shown by Proposition 1, admits no pure-symmetric equilibria for one-round voting, whereas “informative + sincere” remains viable in the two-round mechanism—implying better performance in that setting. In the close distribution, it is difficult to distinguish the world state given a small number of agents, and we expect neither mechanism have high performance.

We run 50 votes for each set of parameters. In each vote, the world state and the signal of every agent is firstly generated from the distribution. Then, for this set of world state and signals, we run a one-round voting and a two-round voting separately. We create n generative AI agents with the signal generated for one-round voting and create another n generative AI agents with the signal generated for two-round voting. For the one-round voting, each agent is asked once for their vote. For the two-round voting, each agent is asked twice. We first ask every agent for their first-round vote simultaneously. Then we collect their first-round votes and ask for their second-round votes. In the second round, the outcome of the first round is included in the prompt, and the question and their answers to the first round consist of the history message.

The prompt for AI agents contains the following components.

Scenario. We chose a hiring setting: each AI agent plays the role of a human resource specialist and votes on whether to hire a candidate.

World state and signals. The world state is whether the candidate is qualified, which is not revealed to the agents. Each AI agent is told to have a Good or Bad impression of the candidate, representing its private signal.

Prior and signal distribution. AI agents are informed of the prior and signal distribution in two different ways. In the *precise distribution* setting, every agent is told the exact distribution (P_H, P_L) and $(P_{hH}, P_{hL}, P_{tH}, P_{tL})$. In the *vague distribution* setting, the information is transferred into a vague description via a Deepseek R1 chatbot. For example, the distribution $P_H = 0.5$, $P_{hH} = 0.8$, and $P_{hL} = 0.6$ is converted the following description.

EXAMPLE 5 (VAGUE INFORMATION IN THE PROMPT.). *Before meeting with the candidate, you thought she had a balanced chance to be qualified. If the candidate is qualified, you will likely have a good impression after meeting with her with a high probability well above half. If the candidate is not qualified, you will be less likely, though still moderately probable, to have a good impression at a likelihood distinctly lower than the qualified scenario but remaining above the halfway mark.* □

Types and preferences. Every agent is told their preferences based on their type.

Voting Mechanisms. Every agent is informed that it will participate in either a one-round or a two-round voting procedure. For the two-round voting, we clarify that the first round does not determine the winner but is publicly observable.

Vote. In both the one-round voting and each round of the two-round voting, every agent is asked to determine and report its best vote (YES or NO), along with a brief explanation of its reasoning.

C.2 Example for answers of an AI agent

Here we give an example of the answer of an agent in the two-round voting. The agent expresses the intention to "share information" in the first round and "update belief" in the second round.

Setting. The Generative AI model is Deepseek R1. The biased signal distribution is employed, but agents receive vague information about signals. Specifically, this agent receives a GOOD impression as a private signal.

First-round Vote. Y.

First-round Answer. To determine the optimal first-round vote, consider Bayesian updating: a GOOD impression strongly indicates the candidate is qualified (91% probability). The first-round vote count informs the second-round voters. Voting Y truthfully contributes to an accurate signal, aiding rational voters in updating their beliefs. Extremists' votes are fixed, but their impact is diluted if rational voters coordinate via the first-round tally. Truthful voting maximizes collective accuracy.

First-round Outcome. 14 of 40 agents vote for Y in the first round, including this agent.

Second-round Vote. N.

Second-round Answer. After observing 14/40 first-round YES votes, Bayesian updating suggests the candidate is likely unqualified. Given rational voters' independent impressions, a low YES count aligns more with an unqualified candidate (expected 8-12 YES) than qualified (28-32). Despite my GOOD impression, the aggregate signal outweighs individual likelihood due to extreme unlikelihood of 14 YES under qualification. Vote NO to align with probable unqualification.